JP – 769

III Semester M.C.A. (Two Years Course) Examination, May/June 2023
(CBCS) (2021 – 22)
COMPUTER SCIENCE
3MCA2 : Quantitative, Teaching and Research Aptitude (Soft Core)

Time : 3 Hours Max. Marks : 70

Instruction : *Answer **any five** from Part – **A** and answer **any four** from Part – **B**.*

PART – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. Of the three numbers, second is twice the first and is also thrice the third. If the average of three numbers is 44. Find the largest numbers.
2. What will be the possibility of drawing a Jack or a Spade from a well shuffled standard deck of 52 playing cards ?
3. Simplify $\frac{1}{\log_3 60} + \frac{1}{\log_4 60} + \frac{1}{\log_5 60}$.
4. Which day of the week was 18th April 1963 ?
5. At what time between 3'O clock and 4'O clock the hands of the clock will be together ?
6. Two pipe A and B can fill a tank in 4 hrs. and 5 hrs. respectively. If they opened on alternate hours and if pipe B is opened first. In how many hours the tank is full ?
7. Define the term teaching. Explain the nature and objectives of teaching in brief.
8. Briefly discuss the research methodologies and its types.

PART – B

Answer **any four** questions. **Each** question carries **ten** marks. **(4×10=40)**

9. a) In how many ways can we arrange the word "FUZZTONE" so that all the vowels come together ? **5**

   b) 50 Labourers can dig a pond in 16 days. How many labourers will be required to dig another pond in 20 days, which is double in size ? **5**

P.T.O.

10. a) 12 years ago, age of P was 3 times the age of Q. After 12 years, ratio of ages to Q to P will be 2 : 3. What is the present age of P ? **5**

    b) At what dates of April 2001 did wednesday fall ? **5**

11. a) Explain the characteristic and types of communication. **5**

    b) Difference between formal education and distance education in India. **5**

12. a) Explain governance policy and administration of higher education system in India. **5**

    b) Define the following terms. **5**

    i) Article

    ii) Workshop.

13. a) A can do $\frac{1}{2}$ of a piece of work in 5 days, B can do $\frac{3}{5}$ of the same work in 9 days and C can do $\frac{2}{3}$ of that work in 8 days. In how many days can all of them together do the work ? **5**

    b) At what time between 7'O clock and 8'O clock the hands of the clock be in opposite direction. **5**

14. a) The compound interest on Rs. 30,000 at 7% per annum for the certain time is Rs. 4,347. Find the time taken in years. **5**

    b) Ajay started a business investing Rs. 12,000. After 5 months, Bala joined him with a capital of Rs. 9,000. At the end of the year the total profit was Rs. 4,416. What is the share of profit of Ajay ? **5**
